

Sri Lanka Institute of Information Technology



IE1030 - Data Communication Networks
Year 1, Semester 1- 2025 June

Network Design

Part 1 and Part 2

Group WD.01.40

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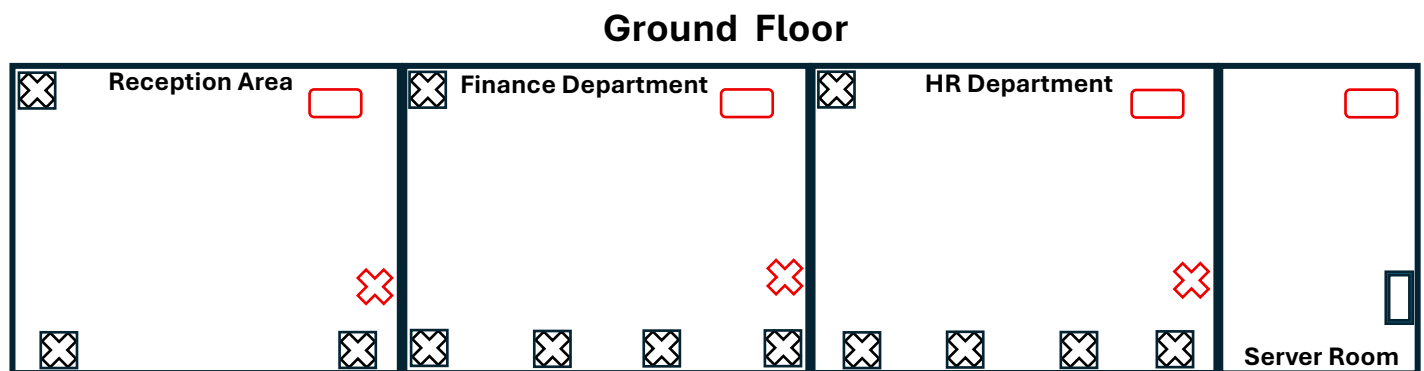
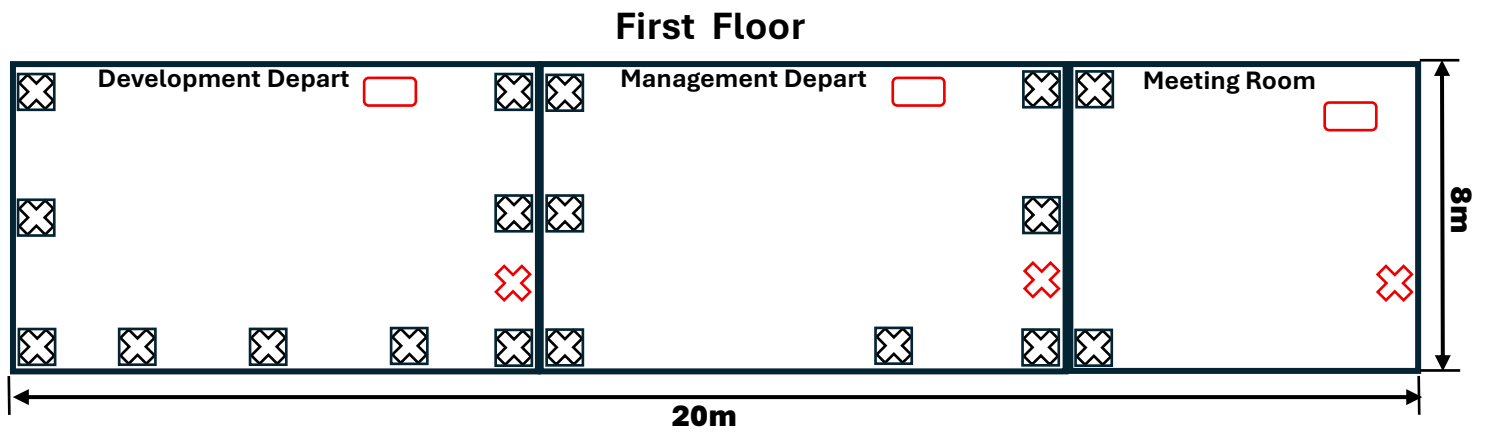
Overview

Our Company currently holds Reception Area, 4 Department, Server Room and Meeting Room

1. Reception Area - 4 devices
2. Finance Department - 6 devices
3. HR Department - 6 devices
4. Server Room - 3 devices
5. Development Department - 10 devices
6. Management Department - 8 devices
7. Meeting Room - 2 devices

1. Requirement Analysis

a) Floor Plan



-  Data Outlet
-  Switch
-  AP
-  Equipment Rack

Scale = 1:170

b) Wired Connectivity

Ground Floor

Reception Area - 4 Devices + 1 Switch

Finance Department - 6 Devices + 1 Switch

HR Department - 6 Devices + 1 Switch

Server Room - 1 Server + Router + 1 Main Switch + 1 AP

First Floor

Development Department - 10 Devices

Management Department - 8 Devices

Meeting Room - 2 Devices

c) Wireless Connectivity

Ground Floor ~ 16 wireless client

First Floor ~ 16 wireless client

(Each AP Supports ~ 5 or 6 Devices (Laptops, Smartphones, Tablets))

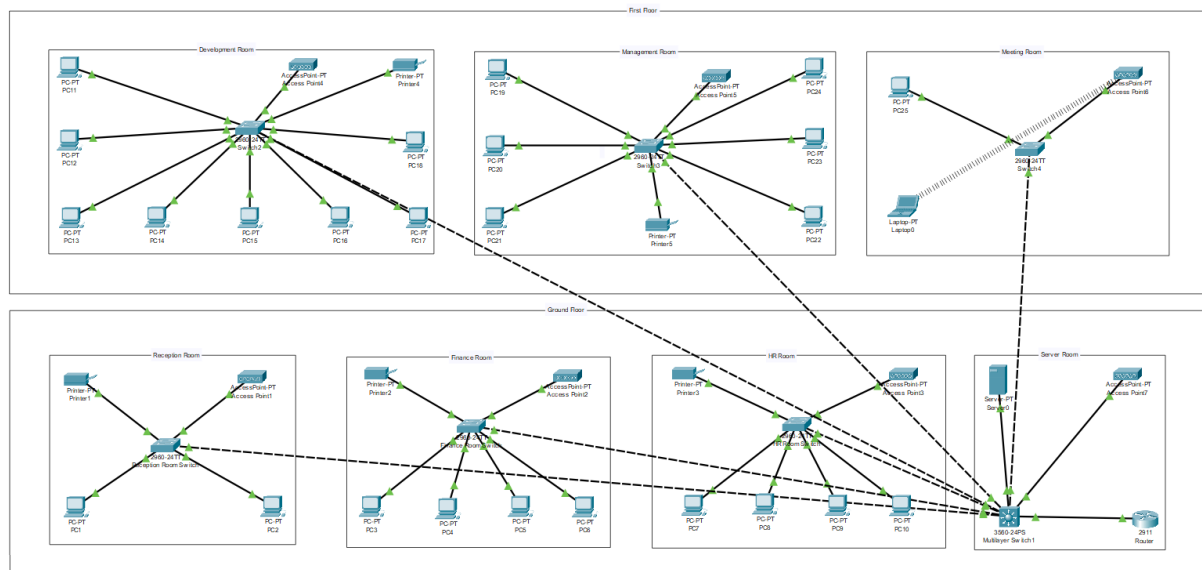
d) Internet connectivity

Type of Connection	Provider	Speed	Plan Detail	Monthly Cost (LKR)
FTTH	SLT	1000 Mbps	Unlimited Data	39,900/month
4G LTE Home Broadband	SLT	10 Mbps	Unlimited Data	7,090/month
ADSL Broadband	SLT	5 Mbps	Unlimited Data	5,790/month

- For this project, we are using an SLT FTTH 1000 Mbps Unlimited connection as the primary network, since FTTH provides high speed, low latency, and stable performance for supporting approximately 70 devices.

2. Logical Network Design

a)

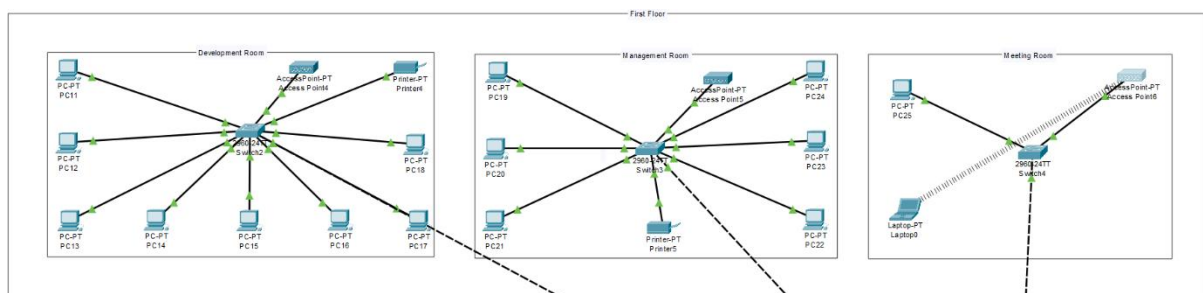


Overview of the network

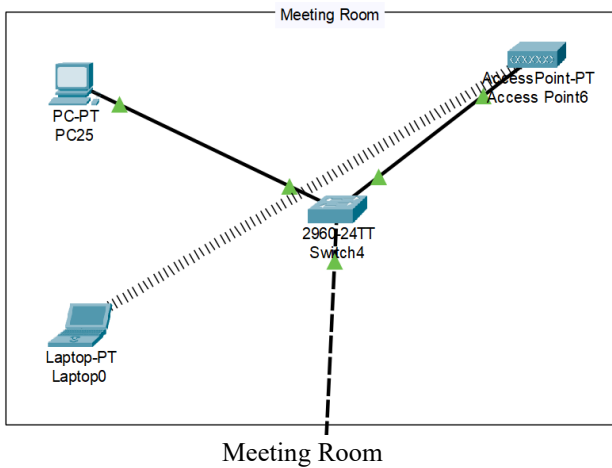
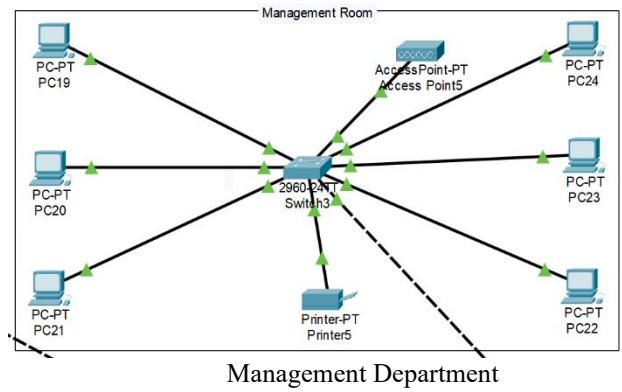
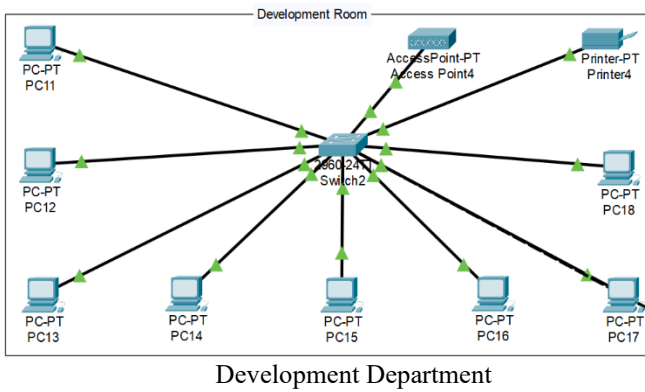
First Floor

The first floor currently contains,

1. Development department
2. Management department
3. Meeting room



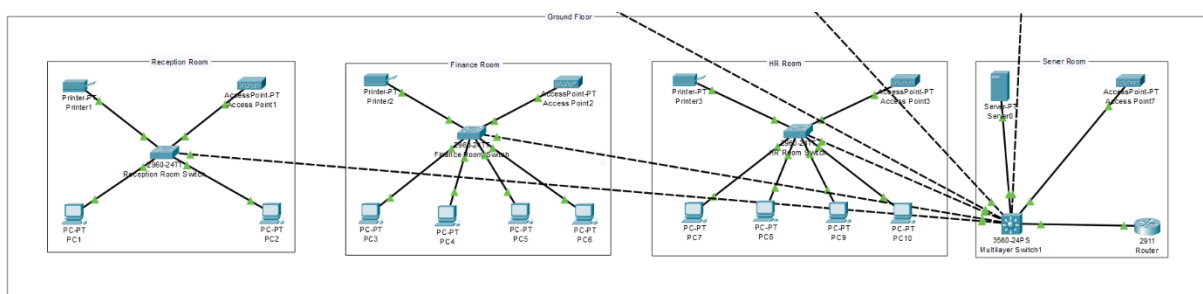
Overview of the First Floor

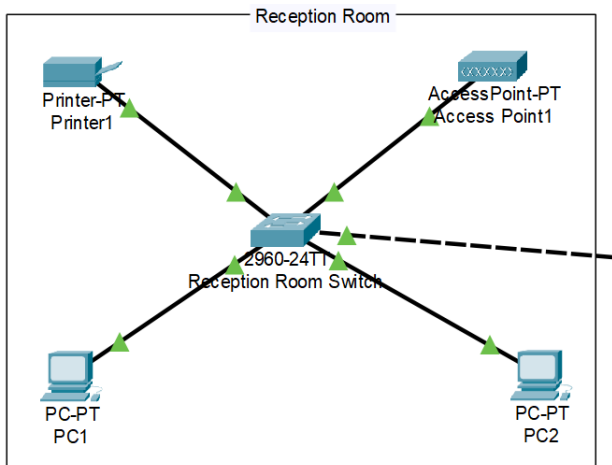


Ground Floor

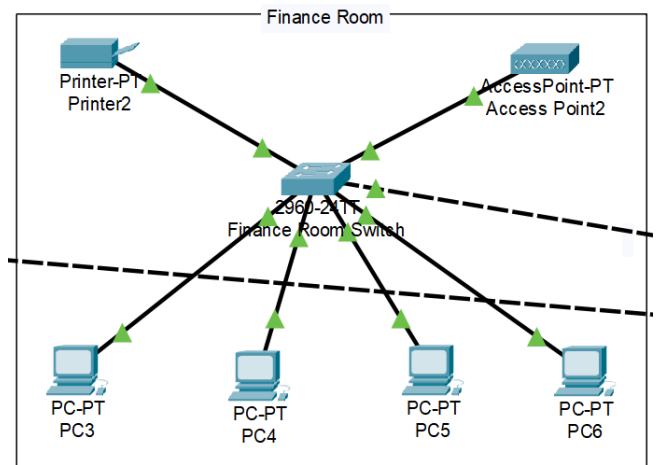
The Ground floor currently contains,

1. Reception Area
2. Finance Department
3. HR Department
4. Server Room

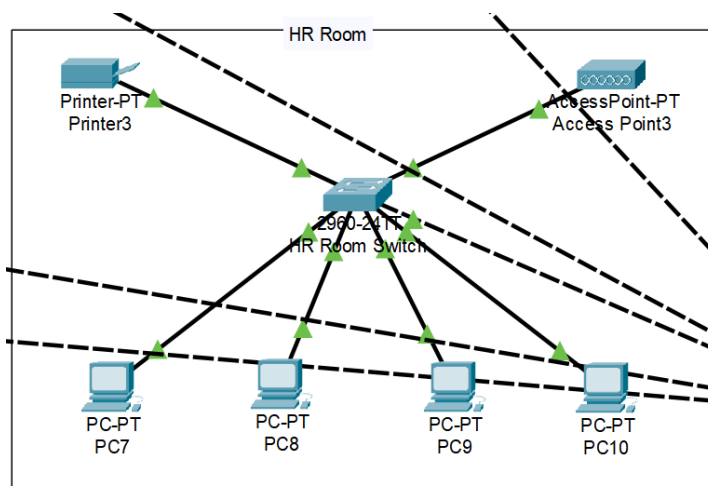




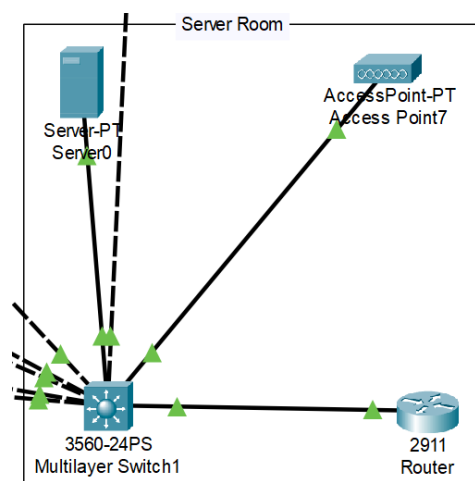
Reception Area



Finance Department



HR Department



Server Room

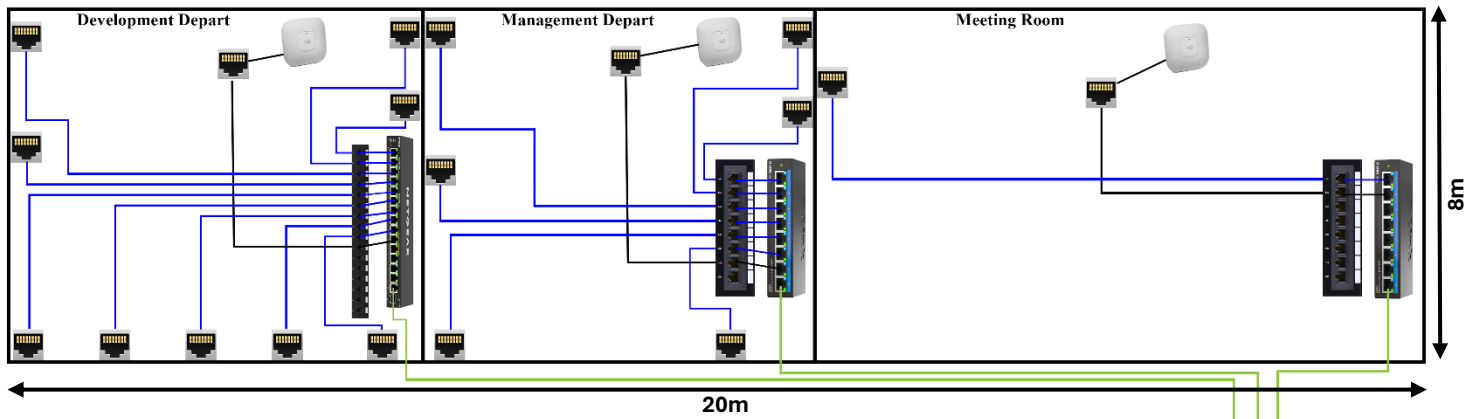
b)

Link	Speed
Hosts - Room Switch	100 Mbps
AP - Room Switch	1 Gbps
Room Switch - Main Switch	1 Gbps
Main Switch - Router	1 Gbps
Main Switch - Server	1 Gbps

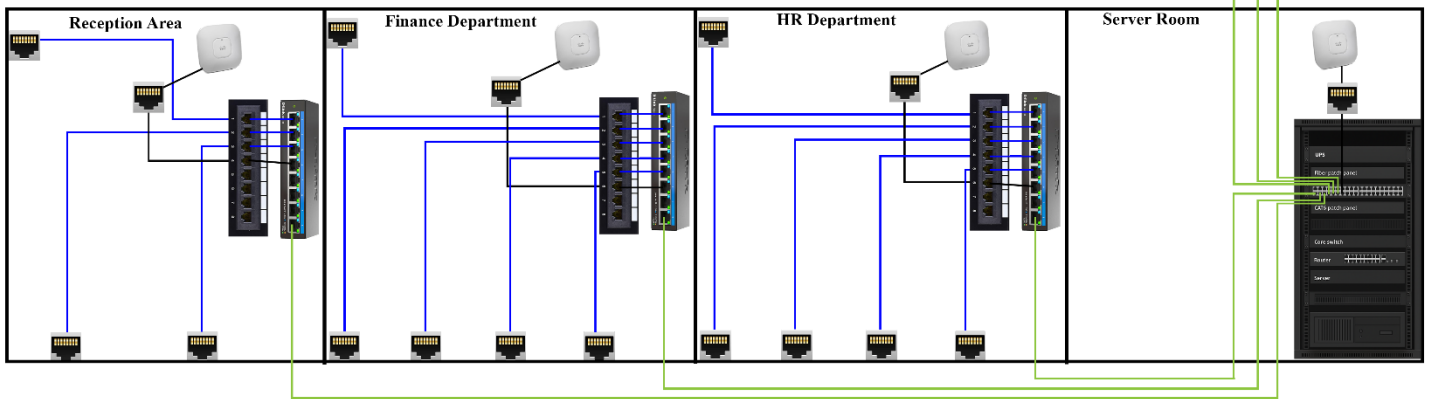
3. Physical network design - Passive Components

Floor Plan

First Floor



Ground Floor



Overview of the Physical Network Design Floor plan



Wall outlet



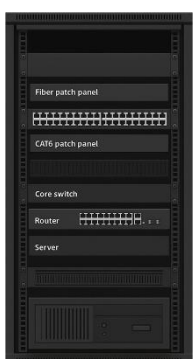
Switch



Patch Panel



AP



Floor Standing Equipment Rack (22U)

a) UTP Patch panels

Overview

The physical layer of the network is designed to provide reliable LAN connectivity using UTP (Unshielded Twisted Pair) cabling. All copper cabling is terminated in patch panels mounted in equipment racks. This design ensures organized cable management, scalability, and ease of maintenance.

- Cable Type - UTP CAT6
- Maximum Segment Length - 90m
- Patch Panel Type - 8 port, 16 port, 24 port (depending on room device count)

Allocation:

First Floor

Departments	Patch Panel Port Requirement
Development Department	16 port patch panel
Management Department	8 port patch panel
Meeting Room	8 port patch panel

Ground Floor

Departments	Patch Panel Port Requirement
Reception Area	8 port patch panel
Finance Department	8 port patch panel
HR Department	8 port patch panel
Server Room	24 port patch panel

- Extra ports are included for future expansion.

b) Wall outlets

Overview

Wall outlets are the termination points for UTP horizontal cabling in each office. They provide a user-friendly interface for connecting end devices such as PCs, printers, and access points to the LAN.

Each wall outlet is connected via CAT6 UTP cable to the nearest patch panel, ensuring structured cabling, easy troubleshooting, and flexibility for reassigning devices.

- Outlet Type - RJ-45 modular jack, CAT6
- Port Configuration- Single
- Installation Height - 30 - 40cm from floor level

Allocation:

First Floor

Departments	Wall Outlets
Development Department	10 outlets
Management Department	8 outlets
Meeting Room	2 outlets

Ground Floor

Departments	Patch Panel Port Requirement
Reception Area	4 outlets
Finance Department	6 outlets
HR Department	6 outlets
Server Room	2 outlets

c) Equipment Rack

Overview

Equipment racks provide a structured housing for passive and active networking components such as patch panels, switches and UPS units. They ensure proper organization, security, ventilation, and accessibility of network equipment.

- Rack Type - Floor Standing Rack (22U)
- Location - Server Room

Height - 2

4. Physical network design - active components

a. Switches

1. Main / Core Switch

- High-capacity, enterprise-grade switch.
- Handles uplinks from all access switches, server traffic, and router connectivity.
- Must support advanced management features, high PoE budget, and fiber uplinks.

Specifications:

- Model - Ubiquiti UniFi Switch Enterprise 24 PoE
- Ports - 24 × RJ45
- PoE - 400W budget, supports 802.3af/at/bt
- Switching Capacity - 124Gbps
- Forwarding Rate - 92.8 Mbps
- Management - UniFi Controller
- Layer Support - L2 + static L3 routing, Inter VLAN routing

2. Access Switches

- Smaller switches dedicated to each office/room.
- Provide direct connections to PCs, printers, and APs.
- Uplink back to the core switch using copper/fiber.
- Must support PoE for powering room wireless APs.

Development Room Access Switches Specifications:

- Model - TP-Link JetStream TL-SG3424P
- Ports - 16 × RJ45
- PoE - 150W budget
- Switching Capacity - 32 Gbps

Other Rooms Access Switches Specifications:

- Model - UniFi Switch Lite-8 PoE
- Ports - 8 × RJ45
- PoE - 52W budget
- Switching Capacity - 16 Gbps
- Forwarding Rate - 11.9 Mbps
- Management - UniFi Controller

b. Access points

- Model - Ubiquiti UniFi 6 Lite (U6-Lite)
- WiFi Standard - IEEE 802.11ax (Wifi 6)
- Frequency Bands - 2.4 GHz, 5 GHz
- Total Throughput - 1.5 Gbps per AP
- Coverage Radius - ~25-30m
- Ethernet Output - 1 × Gigabit Ethernet
- Security - WPA2/WPA3, VLAN tagging, Guest Portal isolation
- Management - UniFi Network Controller
- Mounting - Ceiling mount

c. Uninterruptible power supplies

- Model - APC Smart-UPS C 2000VA (SMC2000I)
- Technology - Line-Interactive, Pure Sine Wave Output
- Capacity - 2000 VA / 1300 W
- Input Voltage - 230V AC
- Output Voltage - 230V $\pm$ 5% regulated
- Battery Runtime - ~7-10 minutes at full load (1300 W)
~20-25 minutes at 50% load (650 W)
- Battery Type: Sealed, Maintenance-Free Lead Acid (VRLA)
- Recharge Time: 3-4 hours to 90% capacity
- Management - USB & Serial ports for monitoring
APC PowerChute software for automatic server shutdown
- Protection - Surge & spike protection
Overload & short-circuit protection
AVR for voltage stabilization

5. Cost estimate

- **Active Components**

Switches, Access Point, Uninterruptible Power Supplies (UPS)

Items	Model	Quantity	Unit Price	Total (LKR)
Main / Core Switch	Ubiquiti UniFi Switch Enterprise 24 PoE	1	236,000	236,000
Access Switch	TP-Link JetStream TL-SG3424P	1	129,500	129,500
Access Switches	UniFi Switch Lite-8 PoE	5	55,500	277,500
Access Point	Ubiquiti UniFi 6 Lite (U6-Lite)	7	36,000	252,000
UPS	APC Smart-UPS C 2000VA (SMC2000I)	1	200,000	200,000
				1,095,000

Cable Installation = LKR 90/m

Total Cable Installation Cost = $610 \times 90 =$ LKR 54,900

Additional Consumables = ~ LKR 10,000

Total Labor Cost = LKR 64,900

- **Passive Components**

UTP cabling, UTP patch cords, Equipment Rack

Items	Model	Quantity	Unit Price	Total (LKR)
UTP Cabling	CAT6, 305m box	2	44,000	88,000
Patch Panels	CAT6 (8,16,24 port mix)	7	15,000	105,000
Wall Outlets	RJ-45 modular jack	36	2,200	79,200
Patch Cords	CAT6, 1-2m	40	800	32,000
Equipment Rack	22U Floor Standing	1	93,000	93,000
				397,200

- **Internet connectivity subscription (per month)**

Type of Connection	Plan Detail	Monthly Cost
FTTH (SLT)	Unlimited Data (1000 Mbps)	LKR 39,900

Total Cost:

Active Components = LKR 1,095,000

Passive Components = LKR 397,200

Labour Cost = LKR 64,900

Internet Connectivity = LKR 39,900

Total Cost = LKR 1,597,000